

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

November 2012

EC 204 Electronics Instruments & Measurement

Date: 30.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) State two types of systematic errors giving example of each. [03]
(b) Give the comparison between LED and LCD [04]
- Q - 2 (a) Why a delay line used in the vertical section of Oscilloscope? Draw and explain delay line circuit. [07]
(b) Explain frequency synthesized signal generator in detail. [07]

OR

- Q - 2 (a) Draw and explain block diagram of Digital Storage Oscilloscope. [07]
(b) Explain Function Generator in detail with block diagram [07]
- Q - 3 (a) Describe the operation of Wheatstone bridge and derive balance equation for resistance measurement. [05]
(b) Find the series equivalent inductance and resistance of the network that causes an opposite angle (Hay bridge) to null with the following bridge arms. (Refer Figure A) [05]

$$\omega = 3000 \text{ rad/s}, R_2 = 10 \text{ k}\Omega, R_1 = 2 \text{ k}\Omega, C_1 = 1 \mu\text{F}, R_3 = 1 \text{ k}\Omega$$

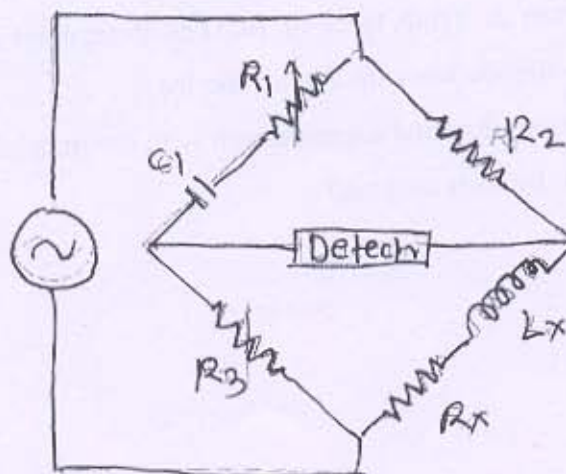


Fig- A

- (c) Explain thermocouple in brief [04]

OR

- Q - 3 (a) Explain how the inductance is measured in terms of known capacitance using Maxwell's bridge. [05]
- (b) Define transducer. Discuss parameters for selection of any transducer. [05]
- (c) What is the balance condition of Wheatstone bridge? A Wheatstone bridge consists of following parameters. $R_1=10\text{ k}\Omega$, $R_2=15\text{ k}\Omega$ and $R_3=40\text{ k}\Omega$. Find unknown resistance R_x [04]

SECTION – II

- Q - 4 (a) Define: (i) Rise time (ii) fall time (iii) Accuracy (iv) Resolution [04]
- (b) Give the difference between active transducer and passive transducer [03]
- Q - 5 (a) Explain construction and operation of LVDT in detail [07]
- (b) What is DAS? Explain with help of a block diagram. [07]

OR

- Q - 5 (a) What do you mean by temperature transducer? Explain Thermocouple and Thermister. [07]
- (b) What are the basic components of a digital system? Explain the operational principle of digital time measurement with block diagram. [07]
- Q - 6 (a) Explain BCD to dot matrix converter in detail. [06]
- (b) Explain the basic circuit for measurement of frequency showing Gate Control F/F. [06]
- (c) What do you mean by seebeck effect? [02]

OR

- Q - 6 (a) Draw logic diagram & Truth table of BCD to 7-segment converter and explain it. Also show its pin connection with display device. [06]
- (b) Explain the operation of period measurement with the multiple average mode. [06]
- (c) What do you mean by over ranging? [02]
